



Department of Electronics and Communication Engineering

Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. ECE B

Course Code & Title: EC8392 Digital Electronics

Name of the Faculty member: Mr.D.Gopinath

Innovative Practice Description

- **Unit / Topic:** III / Flip Flops – SR, D, JK and T
- **Course Outcome:** CO: CO3 The students will be able to analyse and design synchronous sequential circuits
- **Topic Learning Outcome:** TLO: Explain the SR, JK, T, D, Master/Slave Flip-flops operation and its Triggering
- **Activity Chosen:** Jigsaw
- **Justification:**

The Jigsaw Strategy is an efficient way to learn the course material in a cooperative learning style. The jigsaw process encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity. Group members must work together as a team to accomplish a common goal; each person depends on all the others.

- **Time Allotted for the Activity:** 40 minutes

- **Details of the Implementation:**

The lessons are divided into subcategories. Students are divided into groups of four or five students. Then each small group would be created with one student receiving one Subcategory of the lesson. For this method, each small group gets the same set of Subcategories. Once individuals have researched their own subcategory, they will meet with individuals from the other small groups with the same topic to better develop their understanding and become experts of the subcategory. Each student would then return to their original group and teach their subcategory to the rest of their small group.

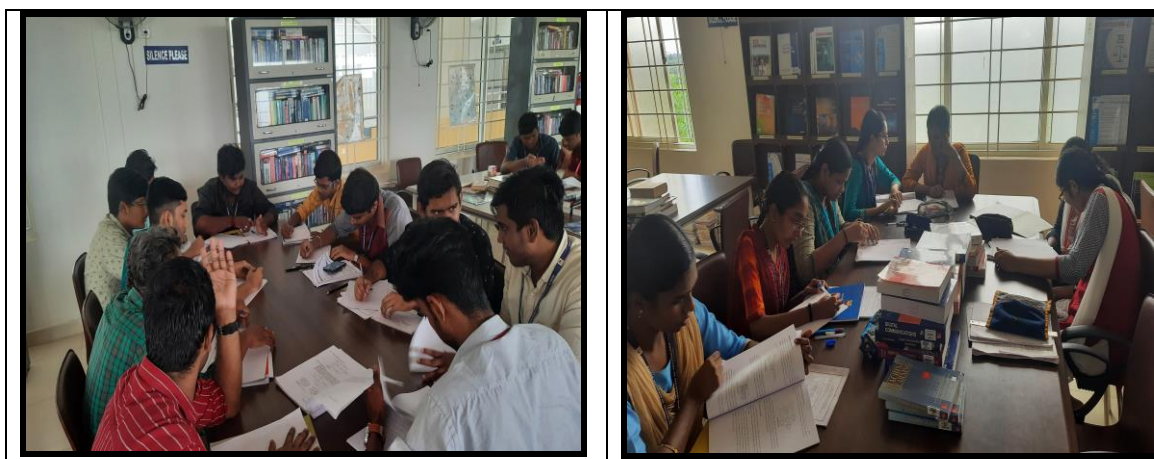
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO8	PO9	PO10	PO12	PSO2	PSO3
CO3	1	2	2	2	2	2	2	2	3

- **PO / PSO mapped:**

Innovative practice	PO8	PO9	PSO3
	2	2	3
Justification for correlation	Identify tenets of the professional code of ethics while doing activity based learning. So, it is mapped at level 2	It helps the students to present results as a team, with smooth integration of contributions from all individual efforts. So, it is mapped at level 2	The concept of memory and programmable logic devices and logic families is required for design, analyze VLSI systems. So it is mapped at level 3

Images / Screenshot of the practice:



- **Reflective Critique:**

❖ ***Feedback of practice from students and other stakeholders:***

Students feel that it was interesting. They learn how to communicate with team members and work together.

❖ ***Benefit of the practice:***

Every student got equal opportunity to come forward to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correct

❖ ***Challenges faced in implementation:***

The main challenge faced is group conflict. Few Students are not ready to work together. To avoid that conflict I started to praise others, taking turns for equal participation, and shared decision making.

References:

- ❖ <https://www.readingrockets.org/strategies/jigsaw>
- ❖ <https://www.teachhub.com/teaching-strategies/2016/10/the-jigsaw-method-teaching-strategy>



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Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. ECE B

Course Code & Title: EC8392 & Digital Electronics

Name of the Faculty member: Mr.D.Gopinath

Innovative Practice Description

- **Unit / Topic:** V/ Memory Devices
- **Course Outcome:** CO: CO5 The students will be able to explain the concepts of memory devices, digital integrated circuits and their characteristics
- **Topic Learning Outcome:** TLO: Explain the different basic memory structure
- **Activity Chosen:** Flipped class room
- **Justification:**

It allows students to learn in their own pace, it encourages students to actively engage with lecture material, it frees up actual class time for more effective, creative and active learning activities and students take control and responsibility for their learning.

- **Time Allotted for the Activity:** 40 minutes

- **Details of the Implementation:**

Specific topic was given to the students learn on their own. Resources like reference book, videos were given to the students. Students are asked to prepare more in depth than before. Students are separate into groups where students are given a task to perform. Get the class back together to share the individual group's work with everyone.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO8	PO9	PO10	PSO2	PSO3
CO3	1	2	2	2	2	2	2	3

- **PO / PSO mapped:**

Innovative practice	PO8	PO9	PSO3
	2	2	3
Justification for correlation	Identify tenets of the professional code of ethics while doing activity based learning. So, it is mapped at level 2	It helps the students to present results as a team, with smooth integration of contributions from all individual efforts. So, it is mapped at level 2	The concept of memory and programmable logic devices and logic families is required for design, analyze VLSI systems. So it is mapped at level 3

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students feel that they have improved self-learning. They learn how to communicate with team members and work together.

- ❖ ***Benefit of the practice:***

Every students got equal opportunity to come forward to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correct

- ❖ ***Challenges faced in implementation:***

The main challenge faced is that few students not exposed to flipped class room. Students struggle with self-discipline and may turn up to class without having absorbed the lesson. To make all students to participate I started to praise others, taking turns for equal participation, and shared decision making.

References:

- ❖ Bergmann, J., & Sams, A. (2012). Flip your classroom: Reach every student in every class every day. Eugene, Or: International Society for Technology in Education.

- ❖ Center for Teaching Innovation at Cornell University. (2017). Flipping the classroom. Retrieved from <https://www.cte.cornell.edu/teaching-ideas/designing-your-course/flipping-the-classroom.html>.



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Department of Electronics and Communication Engineering

Academic Year 2021 – 2022 (Odd Semester)

Active Learning Activity- Unit-I- Digital Fundamentals

Degree, Semester & Branch: III Semester B.E. ECE B

Course Code & Title: EC8392 Digital Electronics

Name of the Faculty member: Mr.D.Gopinath

Name of the Topic	Name of the Activity	Date & Duration
Logic gates, Universal gates	Class Poll	02.09.2021 & 5 Minutes
Karnaugh map Minimization, Don't care conditions	One minute paper	07.09.2021 & 5 Minutes
Quine – McCluskey method of minimization	Think aloud pair problem solving	13.09.2021 & 5 Minutes

Logic gates, Universal gates- Class Poll

DirectPoll

A _____ is a circuit with only one output but can have multiple inputs.

☐ Logic gate
☐ Truth table
☐ Binary circuit
☐ Boolean circuit

VOTE

DirectPoll

1.A _____ is a circuit with only one output ...

☒ Logic gate
☐ Truth table
☐ Binary circuit
☐ Boolean circuit

☐ NOR
☒ NAND

Karnaugh map Minimization, Don't care conditions- One minute paper

One Minute Paper

Karnaugh-Map

Simplify

$F(x,y,z) = \sum(2,3,4,5)$

$F(x,y,z) = \bar{x}'y + xy'$

K-Map.

$F(x,y,z) = \sum(2,3,4,5)$

Solution

$F(x,y,z) = \bar{x}'y + xy'$



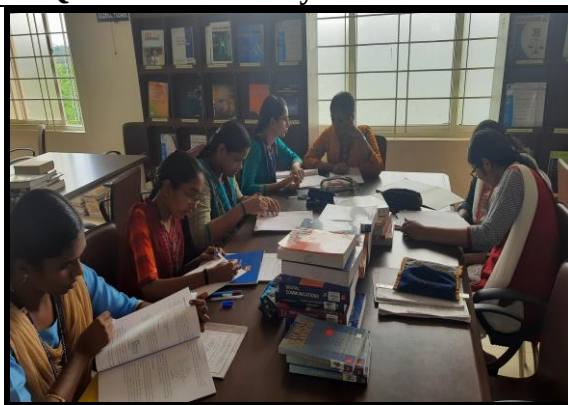
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Quine – McCluskey method of minimization- Think aloud pair problem solving



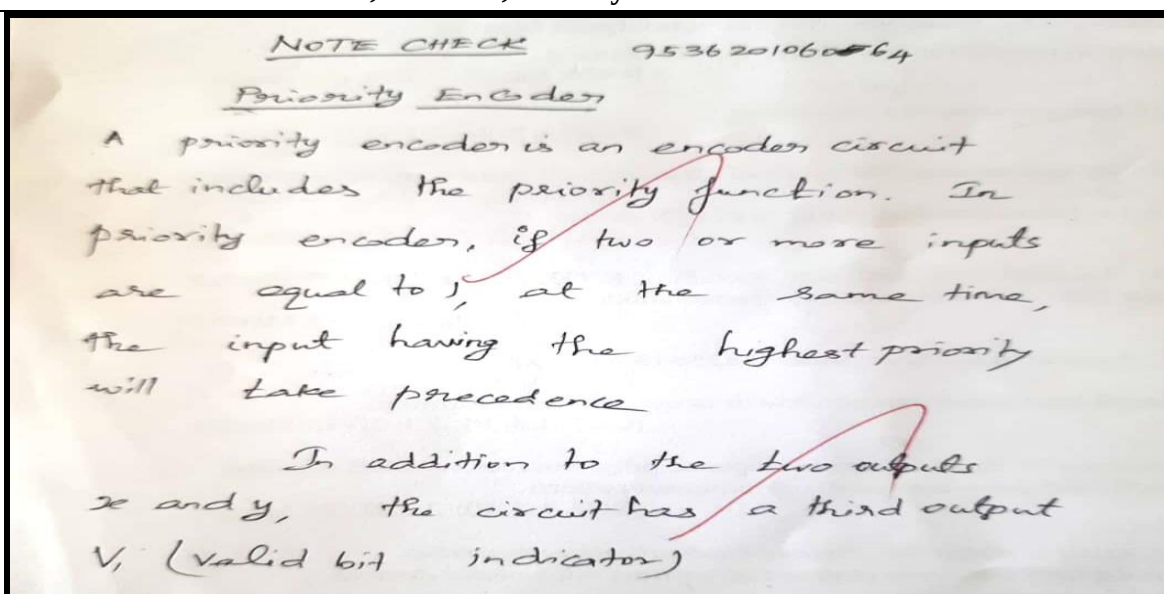
Active Learning Activity- Unit-II- Combinational Circuit Design

Name of the Topic	Name of the Activity	Date & Duration
Multiplexer, Demultiplexer	Open book test	01.10.2021 & 10 Minutes
Decoder, Encoder, Priority Encoder	Note Check	08.10.2021 & 5 Minutes

Multiplexer, Demultiplexer- Open book test



Decoder, Encoder, Priority Encoder- Note Check





Active Learning Activity- Unit-III- Synchronous Sequential Circuits

Name of the Topic	Name of the Activity	Date & Duration
Flip Flops – JK,T	Mind map	11.10.2021 & 5 Minutes
Design – Moore & Mealy models	Zero-Minute Speech	22.10.2021 & 5 Minutes
Design of Counters – Ripple Counters, Ring counters	Reciprocal Peer Questioning	23.10.2021 & 5 Minutes

Flip Flops – JK,T- Mind map

JK Flip Flop

Input J K		Output Q_{n+1}	Action
0	0	Q_n	No change
0	1	0	Reset
1	0	1	Set
1	1	$\bar{Q}_n$	Toggle

PS Q_n	Input J K	NS Q_{n+1}
0	0 0	0
0	0 1	0
0	1 0	1
0	1 1	1
1	0 0	1
1	0 1	0
1	1 0	0
1	1 1	0

Design – Moore & Mealy models- Zero-Minute Speech





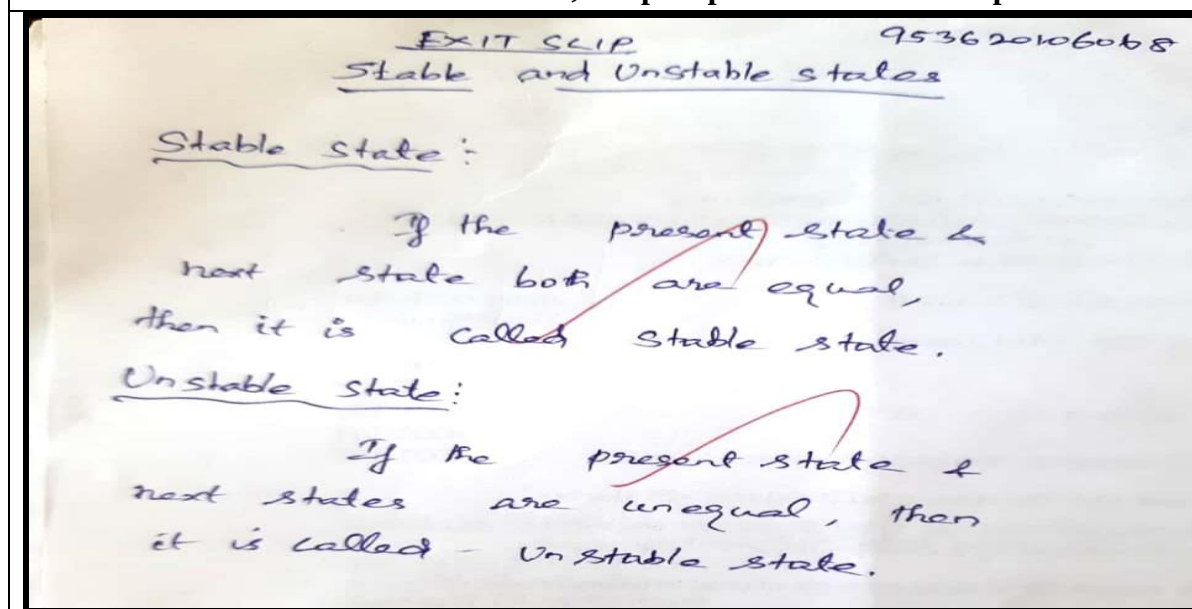
Design of Counters – Ripple Counters, Ring counters- Reciprocal Peer Questioning



Active Learning Activity- Unit-IV- Asynchronous Sequential Circuits

Name of the Topic	Name of the Activity	Date & Duration
Stable and Unstable states, Output specifications	Exit slip	01.11.2021 & 5 Minutes
Hazards, Essential Hazards	Think Pair Share	15.11.2021 & 5 Minutes
Design of Hazard free circuits	Concept mapping	18.11.2021 & 5 Minutes

Stable and Unstable states, Output specifications- Exit slip





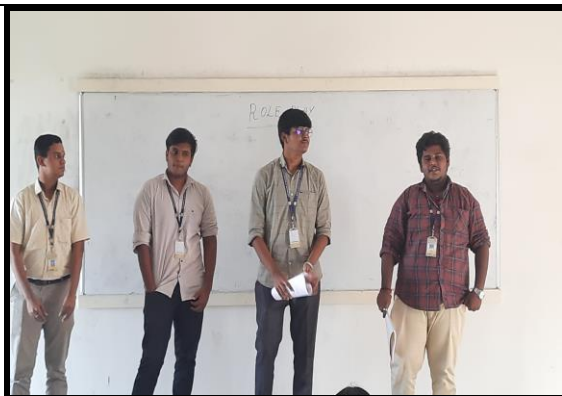
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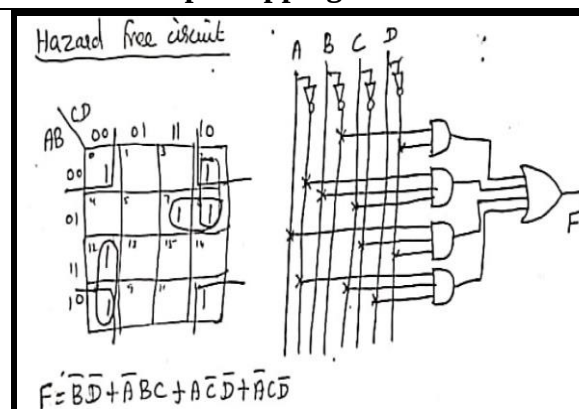
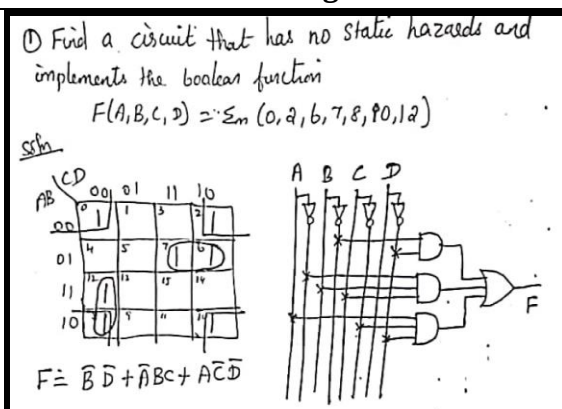
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Hazards, Essential Hazards- Think Pair Share



Design of Hazard free circuits- Concept mapping



Active Learning Activity- Unit-V- Memory Devices and Digital Integrated Circuits

Name of the Topic	Name of the Activity	Date & Duration
PROM, EPROM, EEPROM, EAPROM	Role play	19.11.2021 & 5 Minutes
Implementation of combinational logic circuits using PLA, PAL	Concept Tests	24.11.2021 & 5 Minutes
Logic families and their characteristics – RTL, TTL	Visible Quiz	24.11.2021 & 5 Minutes

PROM, EPROM, EEPROM, EAPROM- Role play

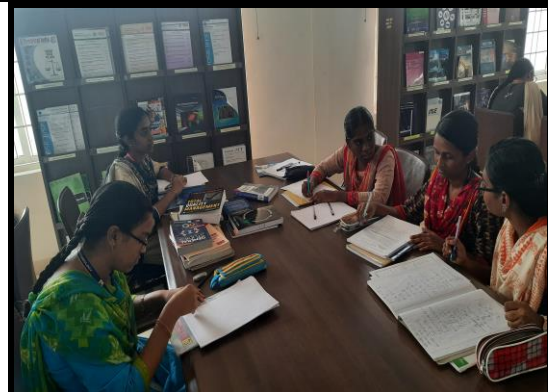
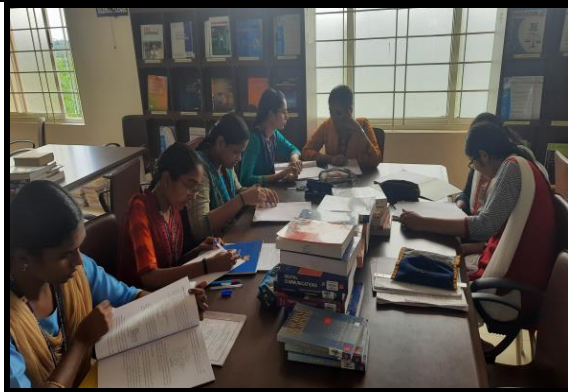




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Implementation of combinational logic circuits using PLA, PAL- Concept Tests



Logic families and their characteristics – RTL, TTL- Visible Quiz

DirectPoll  Help FAQ Education

Poll not starting? How to use?

PAUSE NEXT STOP

1. The fastest saturated logic family is

Questions • Links • Look

Delete Poll Reset Results Save Poll

Questions 0

1

The fastest saturated logic family is

TTL

ECL

IL

MOS

Vote: 0

Settings

Hide answers while polling

Mark the correct answer

Show results in %

QUESTION TYPE

single choice

multiple choice

DirectPoll 

The fastest saturated logic family is

TTL

ECL

IL

MOS

VOTE